BPO: Bayesian Preference Optimization for *Multiobjective Discrete Optimization*

Rahul Patel, **Elias B. Khalil**
University of Toronto

Code + Slides: https://github.com/rahulptel/BPO

Multiobjective Discrete Optimization (MODO)

Applications

Multiobjective Discrete Optimization (MODO)

Applications

Math. Formulation

$$\min_{\mathbf{x} \in \mathcal{X} \subset \mathbb{Z}^N} \Big(f_1(\mathbf{x}),\, f_2(\mathbf{x}), \cdots,\, f_K(\mathbf{x}) \Big)$$

Multiobjective Discrete Optimization (MODO)

Applications

Math. Formulation

$$\min_{\mathbf{x}\in\mathcal{X}\subset\mathbb{Z}^N} \Big(\underline{f_1(\mathbf{x}),\ f_2(\mathbf{x}), \cdots, f_K(\mathbf{x})}\Big)$$

Decision variables

Conflicting objectives

Discrete feasible set

Multiobjective Discrete Optimization (MODO)

Applications

Math. Formulation

$$\min_{\mathbf{x}\in\mathcal{X}\subset\mathbb{Z}^N} \left(f_1(\mathbf{x}),\, f_2(\mathbf{x}), \cdots, f_K(\mathbf{x})\right)$$

Decision variables

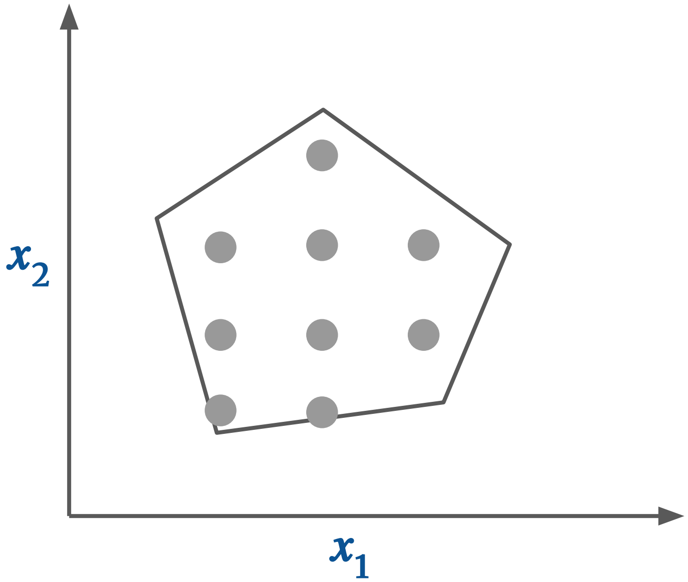
Conflicting objectives

Discrete feasible set

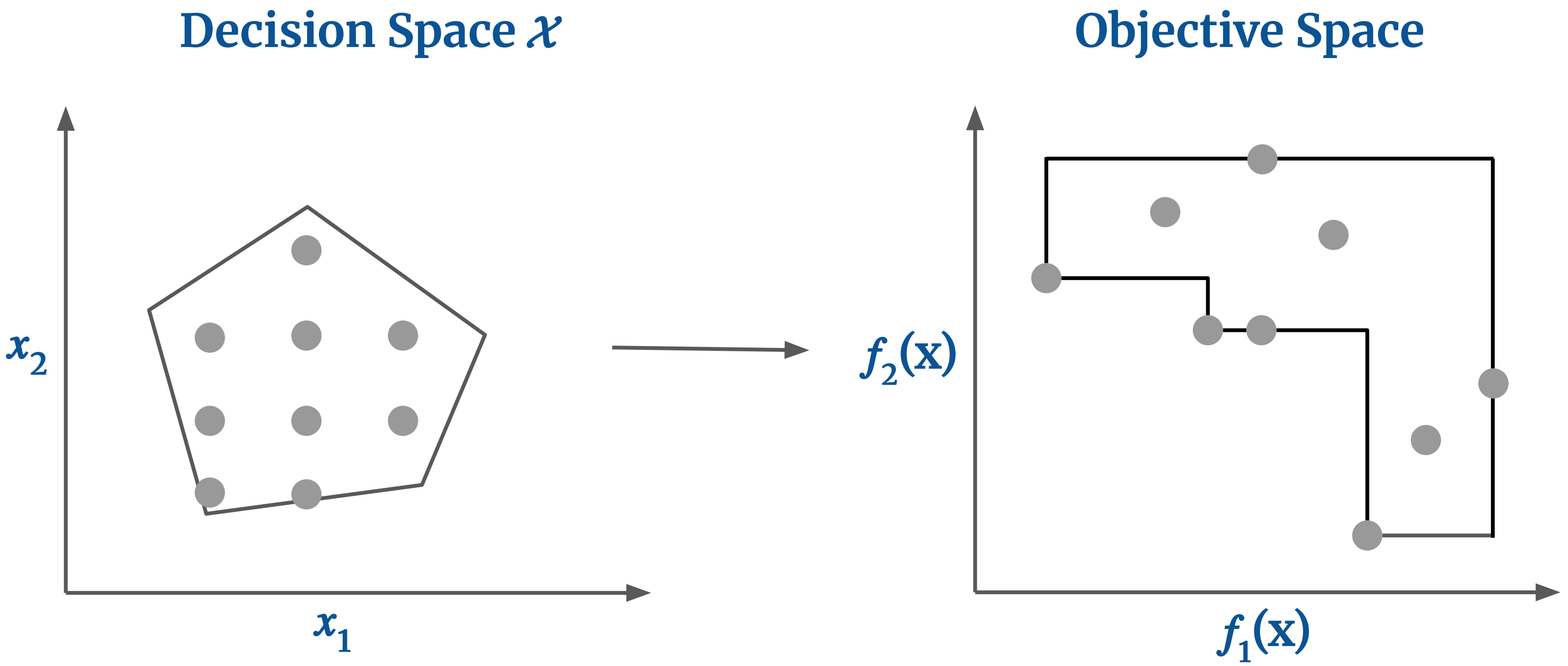
- Solutions represent tradeoffs
- No single winner
- **Goal**: Enumerate Pareto frontier

Decision Space and Objective Space in MODO

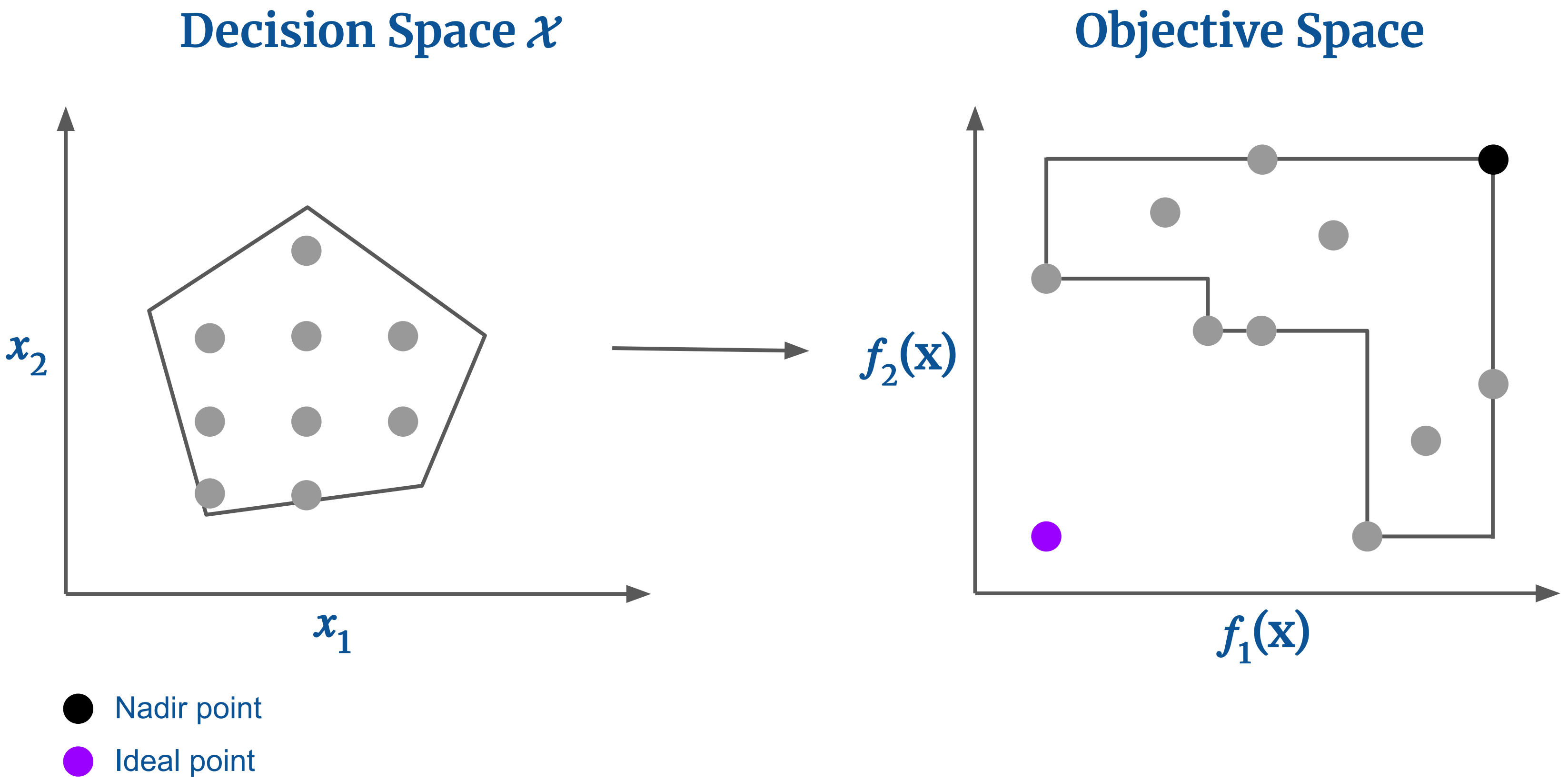
Decision Space $\mathcal{X}$



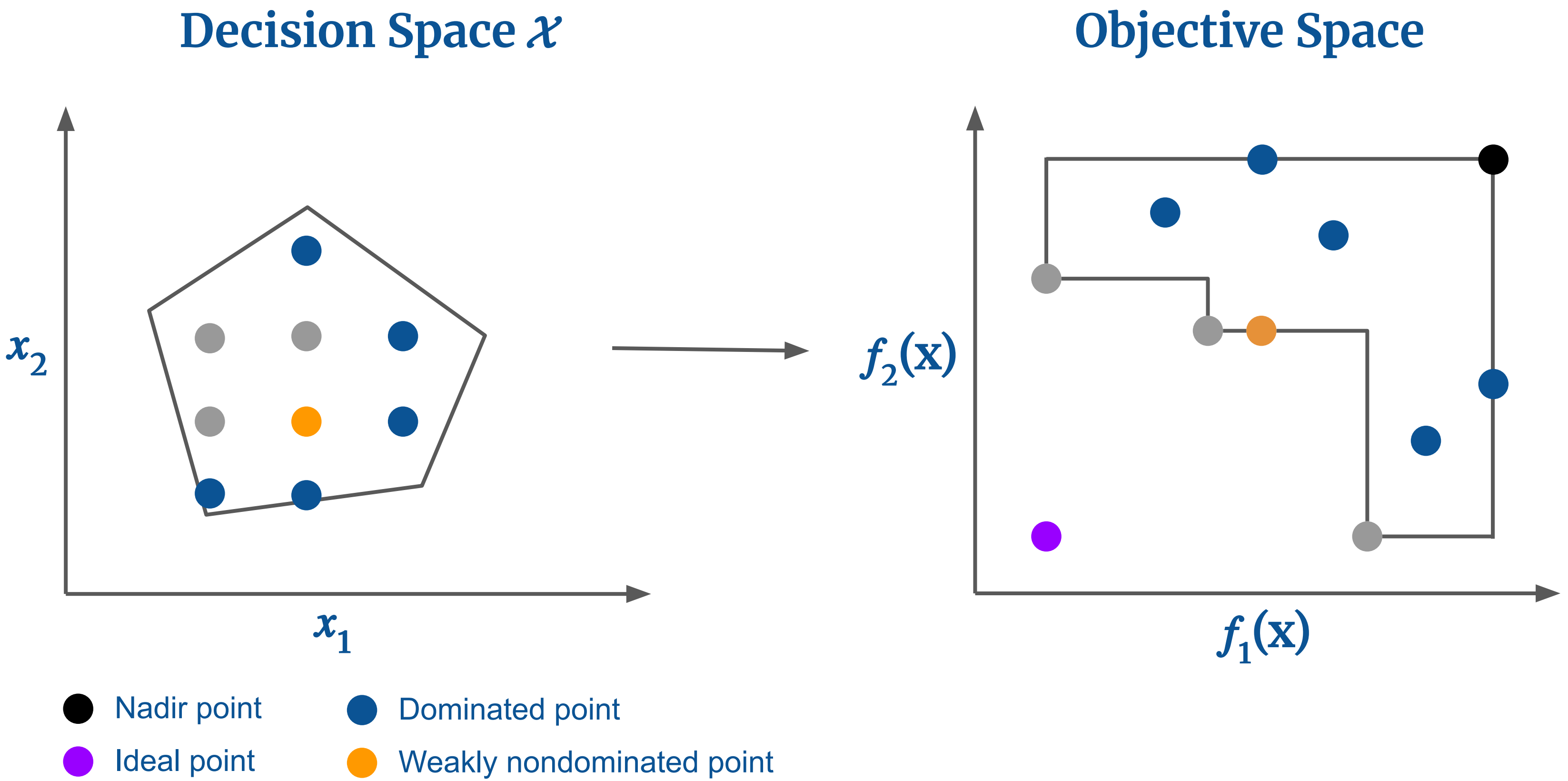
Decision Space and Objective Space in MODO



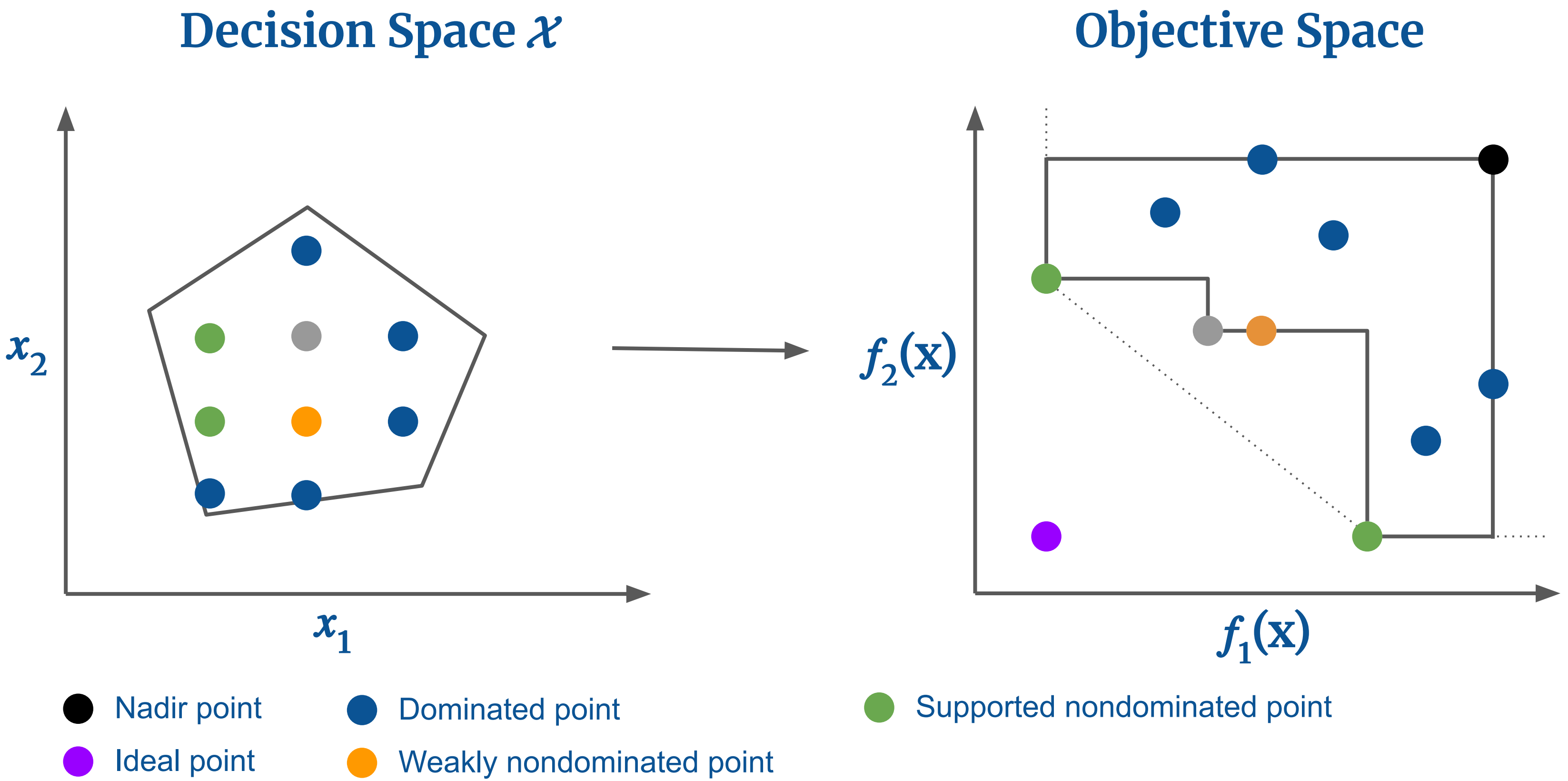
Decision Space and Objective Space in MODO



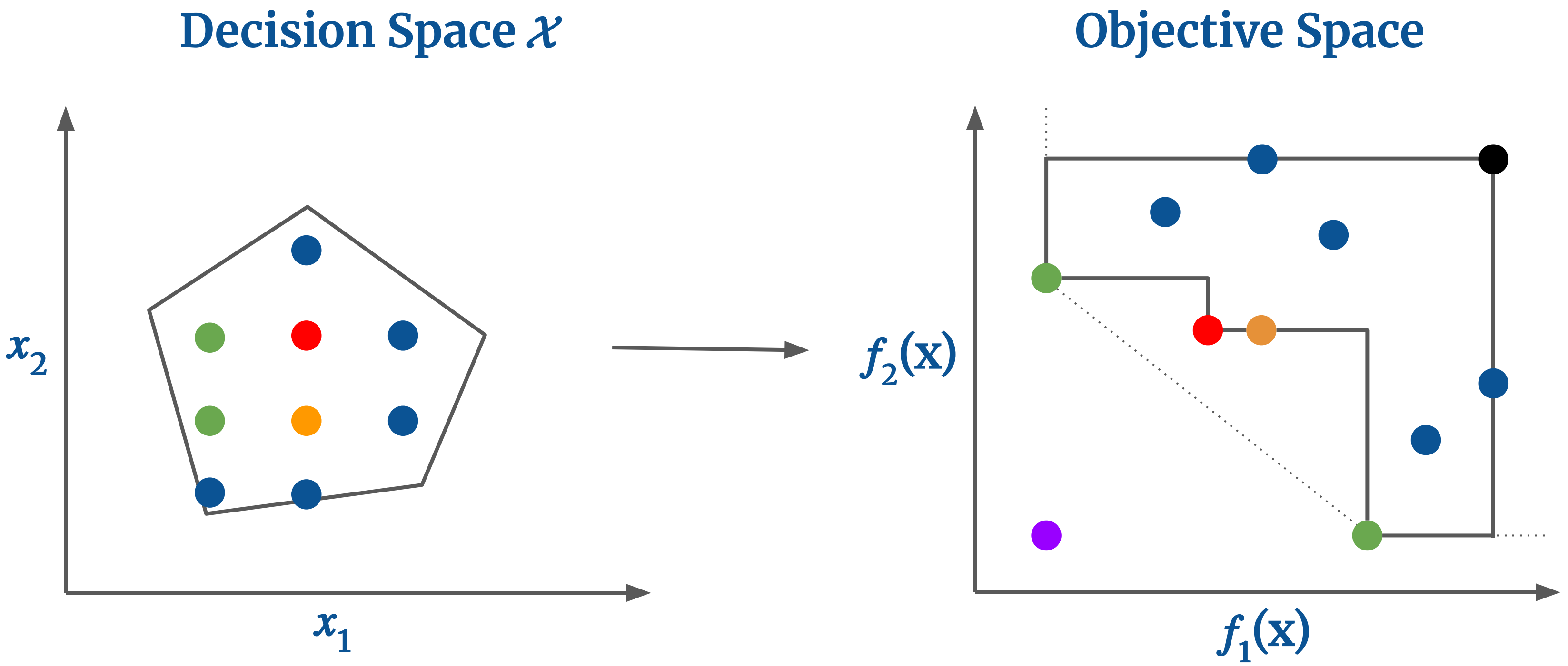
Decision Space and Objective Space in MODO



Decision Space and Objective Space in MODO



Decision Space and Objective Space in MODO



Nadir point
Ideal point
Dominated point
Weakly nondominated point
Supported nondominated point
Unsupported nondominated point

Solution Approaches to MODO problems

Exact Methods

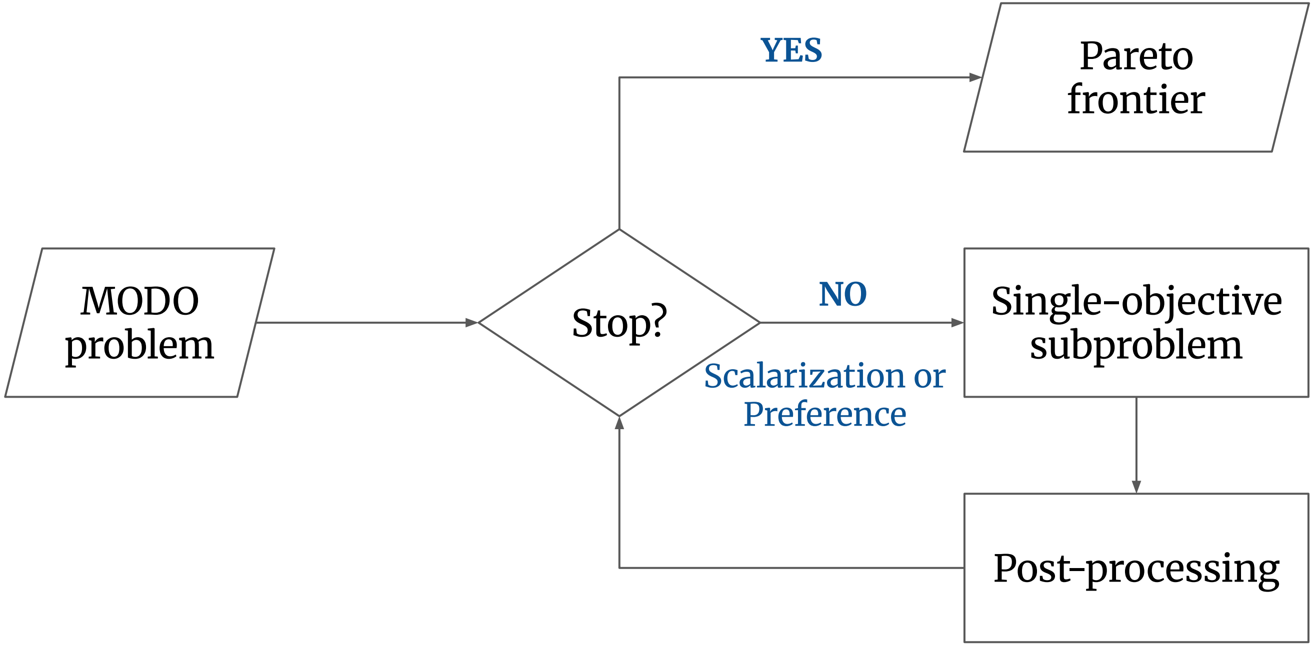
- **Objective Space Search**
  Kirlik and Sayin (2014)
  Boland et al. (2015)
- **Decision Space Search**
  Mavrotas and Danae (1998)
  Adelgren and Gupte (2017)
- **Dynamic Programming**
  Bergman et al. (2022)

Survey on exact MODO approaches:
Halffmann et al. (2022)

Heuristic Methods

- **Matheuristics**
  Pal and Charkhgard (2019)
  An et al. (2024)
- **Evolutionary Algorithms**
  Deb et al. (2002)
  Zhang and Li (2007)
- **Machine Learning**
  Li et al. (2021)
  Lin et al. (2022)

MODO Solution Schematic: A High-Level Overview

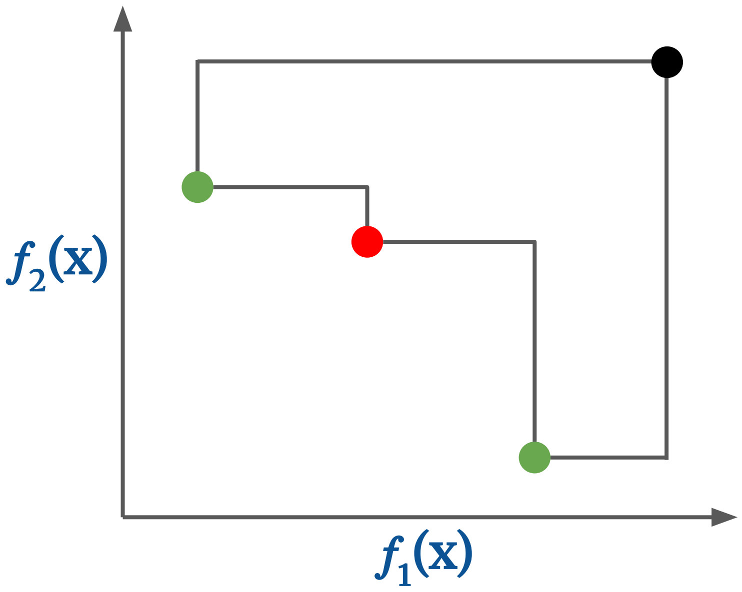


- Prune search space
- Select next preference
- Save solution

Scalarized Subproblem: Weighted-Sum Scalarization

$$\min_{\mathrm{x} \in X} \left(\sum_{i=1}^{m} \lambda_i f_i(\mathrm{x}) \right)$$

$$\lambda_i \geq 0, \quad \sum_{i=1}^{m} \lambda_i = 1$$

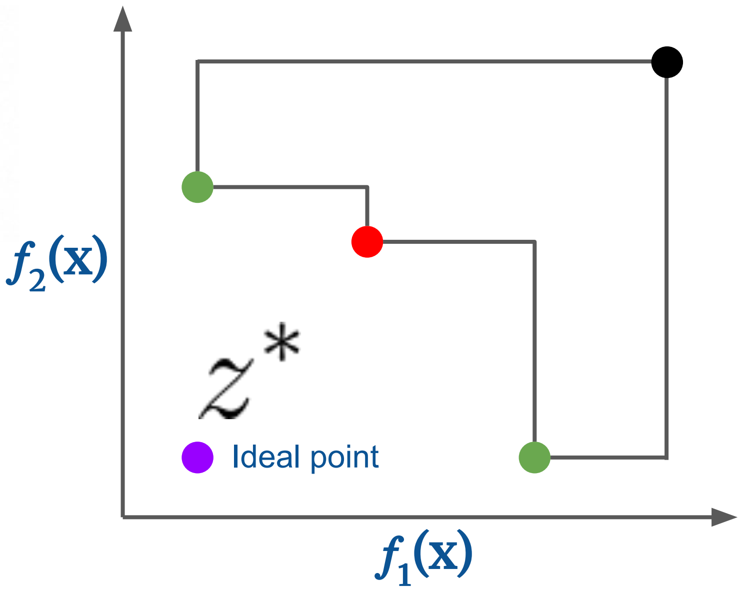


Recovers supported nondominated points but **misses** the unsupported ones

Ehrgott (2005)

Scalarized Subproblem: Augmented Tchebycheff Scalarization (ATS)

$$\min_{\mathrm{x} \in X} \left(\max_{i \in \{1,\ldots,m\}} \lambda_i \big(f_i(\mathrm{x}) - z_i^*\big) + \rho \sum_{i=1}^{m} \lambda_i \big(f_i(\mathrm{x}) - z_i^*\big) \right)$$

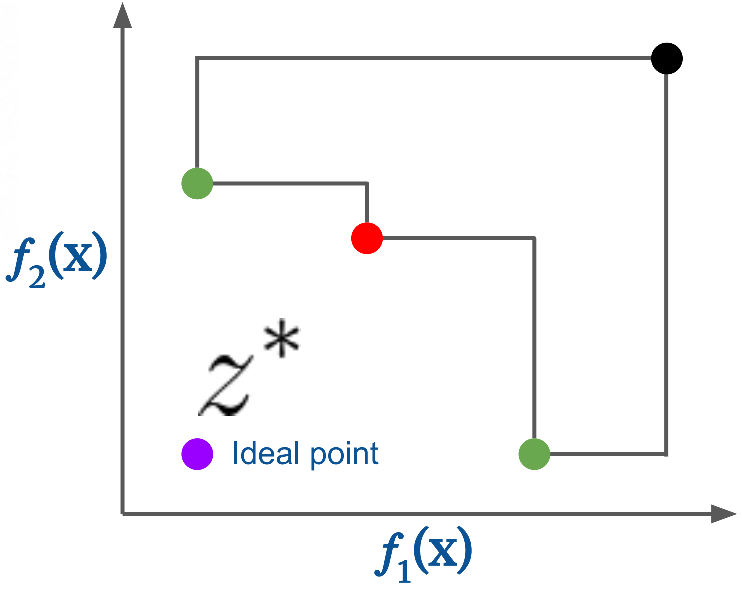


Steuer and Choo (1983)

Scalarized Subproblem: Augmented Tchebycheff Scalarization (ATS)

$$\min_{x \in X} \left(\boxed{\max_{i \in \{1,\ldots,m\}} \lambda_i \left(f_i(x) - z_i^*\right)} + \rho \sum_{i=1}^{m} \lambda_i \left(f_i(x) - z_i^*\right) \right)$$

- Recovers supported and unsupported nondominated points



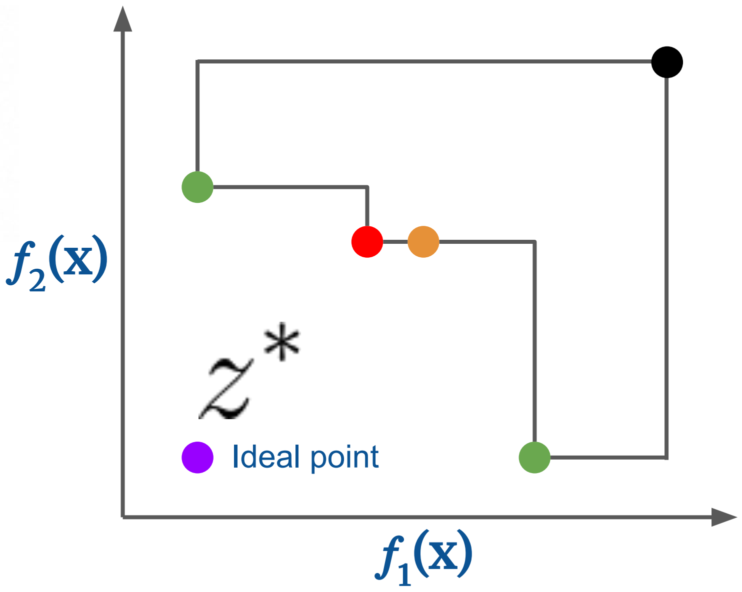
Steuer and Choo (1983)

Scalarized Subproblem: Augmented Tchebycheff Scalarization (ATS)

$$\min_{\mathrm{x} \in X} \left(\max_{i \in \{1,\ldots,m\}} \lambda_i \big(f_i(\mathrm{x}) - z_i^*\big) + \boxed{\rho \sum_{i=1}^{m} \lambda_i \big(f_i(\mathrm{x}) - z_i^*\big)} \right)$$

- Recovers supported and unsupported nondominated points

- Avoids weakly nondominated points



Great explainer: Tripp, Austin. 2025. "Chebyshev Scalarization Explained." May 12. https://austintripp.ca/blog/2025-05-12-chebyshev-scalarization/.

Steuer and Choo (1983)

Limitations

Exact Methods

- Do not scale for large problems
- Focused on enumeration thus could return a subset of the Pareto frontier that lacks diversity

Heuristic Methods

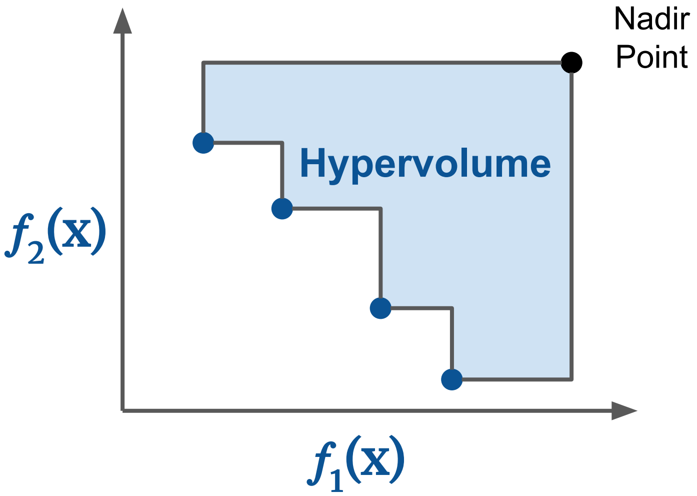
- Struggle to handle hard constraints
- Uniformly sampled preferences
- Static set of preferences

GOAL

Generate high-quality approximate Pareto frontier
as quickly as possible

How to adaptively select preferences in a principled way?

How to Quantify the Goodness of a Preference?



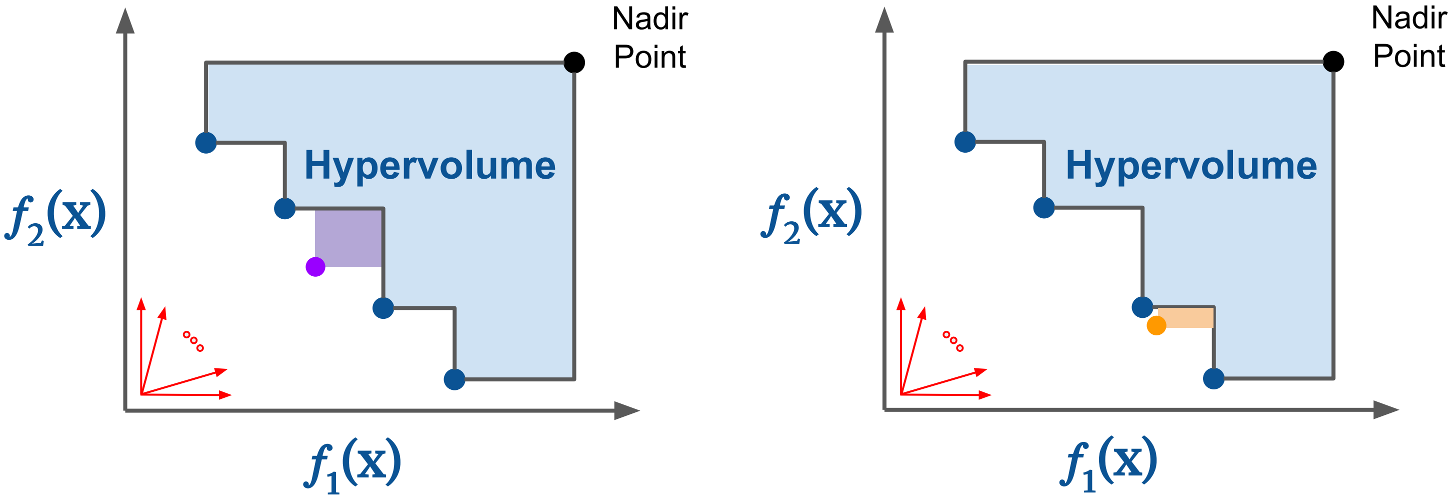
Hypervolume: Volume of the region dominated by non-dominated solutions set relative to a nadir point.

Property: the higher the hypervolume, the better the approximation

Goal

Given **(preference, objective vector)** pairs, select a **new preference** such that the improvement in hypervolume is maximized

How to Select Preferences?



Select a preference that results in the purple point

Preference Optimization as a Bilevel Problem

$$\begin{aligned}
\max_{\mathbf{y},\mathbf{x},\boldsymbol{\lambda}} \quad & \Phi\big(\mathcal{Y}_k \cup \{\mathbf{y}\}\big) \\
\text{s.t.} \quad & \mathbf{y} = f(\mathbf{x}), \\
& \mathbf{x} = \arg\min_{\mathbf{x}' \in \mathcal{X}} \left(\max_{i \in \{1,\dots,m\}} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) \ + \ \rho \sum_{i=1}^{m} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) \right), \\
& \boldsymbol{\lambda} \in \boldsymbol{\Lambda}.
\end{aligned}$$

Preference Optimization as a Bilevel Problem

$$
\begin{aligned}
\max_{\mathbf{y},\mathbf{x},\boldsymbol{\lambda}} \quad & \Phi\big(\mathcal{Y}_k \cup \{\mathbf{y}\}\big) \\
\text{s.t.} \quad & \mathbf{y} = f(\mathbf{x}), \\
& \mathbf{x} = \arg\min_{\mathbf{x}' \in \mathcal{X}} \left(\max_{i \in \{1,\ldots,m\}} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) \; + \; \rho \sum_{i=1}^{m} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) \right), \\
& \boxed{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}.}
\end{aligned}
$$

Select a preference …

Preference Optimization as a Bilevel Problem

$$\begin{aligned}
\max_{\mathbf{y},\mathbf{x},\boldsymbol{\lambda}} \quad & \Phi(\mathcal{Y}_k \cup \{\mathbf{y}\}) \\
\text{s.t.} \quad & \boxed{\begin{aligned} & \mathbf{y} = f(\mathbf{x}), \\ & \mathbf{x} = \arg\min_{\mathbf{x}' \in \mathcal{X}} \left(\max_{i \in \{1,\ldots,m\}} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) + \rho \sum_{i=1}^{m} \lambda_i \big(f_i(\mathbf{x}') - z_i^\star\big) \right), \end{aligned}} \\
& \boldsymbol{\lambda} \in \boldsymbol{\Lambda}.
\end{aligned}$$

Select a preference **with an objective vector** …

Preference Optimization as a Bilevel Problem

$$
\begin{aligned}
\boxed{\max_{\mathbf{y},\mathbf{x},\boldsymbol{\lambda}} \quad \Phi\big(\mathcal{Y}_k \cup \{\mathbf{y}\}\big)} \\
\text{s.t.} \quad & \mathbf{y} = f(\mathbf{x}), \\
& \mathbf{x} = \arg\min_{\mathbf{x}' \in \mathcal{X}} \left(\max_{i \in \{1,\ldots,m\}} \lambda_i\big(f_i(\mathbf{x}') - z_i^\star\big) + \rho \sum_{i=1}^{m} \lambda_i\big(f_i(\mathbf{x}') - z_i^\star\big) \right), \\
& \boldsymbol{\lambda} \in \boldsymbol{\Lambda}.
\end{aligned}
$$

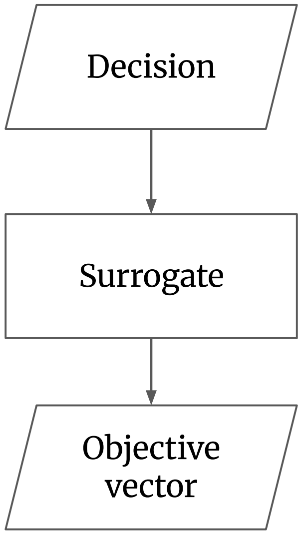
Select a preference with an objective vector **that maximizes the improvement in Pareto frontier quality**

Formulate **next** preference selection as a Bayesian optimization problem

Surrogate Modeling using Gaussian Process

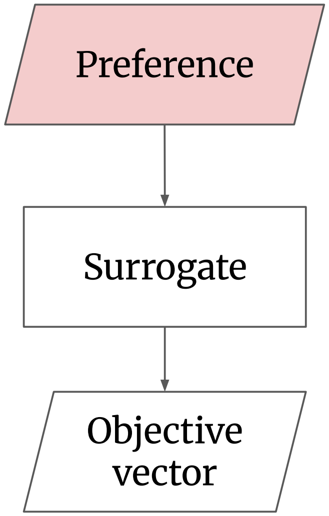
Traditional Multiobjective BO

Assumption: Evaluating a decision is hard



Bayesian Preference Optimization

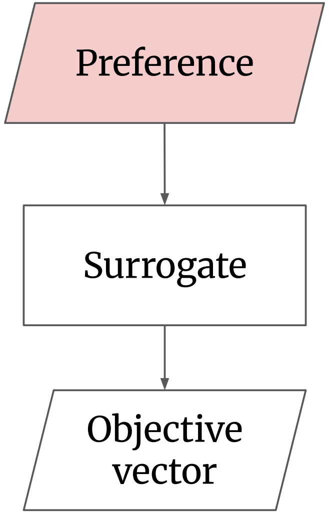
Assumption: Finding good quality decision is hard



Surrogate Modeling using Gaussian Process

Bayesian Preference Optimization

Assumption: Finding good quality decision is hard



Benefits

1. Scalability w.r.t the number of variables
2. Ability to handle mixed-integer variables

Acquisition Function: Expected Hypervolume Improvement

$$\Phi_{\mathrm{qEHVI}} = \mathbb{E}_{f(\Lambda)|\mathcal{D}}\left[\mathrm{HV}(\mathcal{Y} \cup f(\Lambda)) - \mathrm{HV}(\mathcal{Y})\right]$$

- Uses box decomposition of the non-dominated region to compute hypervolume improvement
- Uses a differentiable Monte Carlo estimator, enabling gradient-based acquisition optimization

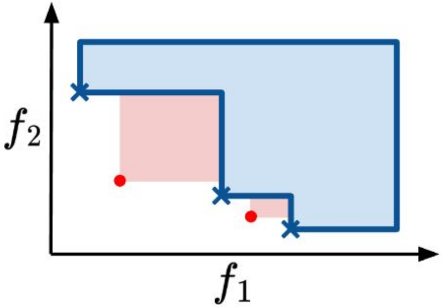
Daulton et al. (2020)

Bayesian Preference Optimization: Active Learning Loop

1. Data Collection

Solve scalarization subproblems

Obtain approximate Pareto Frontier

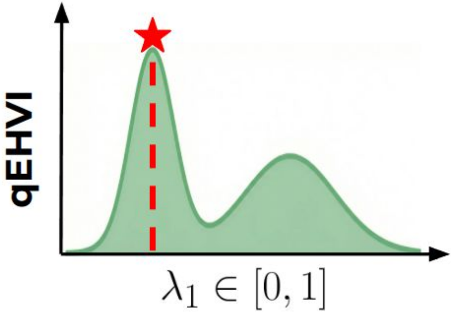


- Dominated Region
- Hypervolume improvement
- Recovered nondominated point (NDP)
- Unrecovered NDP

2. Gaussian Process Surrogate Training

$$p(f(\lambda)|\mathcal{D},\lambda) = \mathcal{N}(\mu_{f(\lambda)}, \sigma_{f(\lambda)})$$

3. Acquisition Optimization



$$\lambda_1 : \lambda_1 + \lambda_2 = 1$$

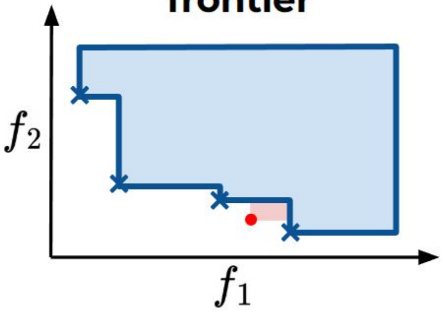
qEHVI peaks when λ_1 = 0.2 preference

4. Solve Subproblem

$$\lambda \longrightarrow \boxed{\text{ATS}} \longrightarrow f(\lambda)$$

Solve ATS with preference optimized using qEHVI

5. Updated Pareto frontier



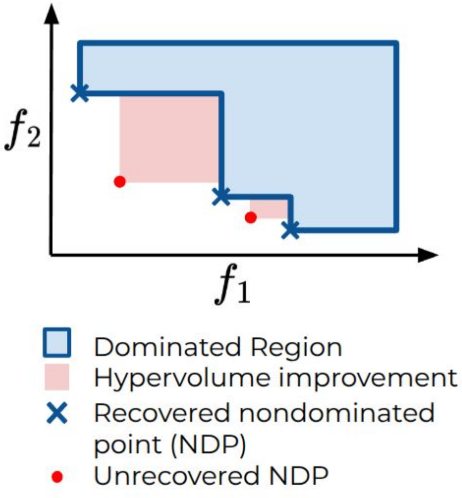
Recovered NDP that maximizes EHVI

Bayesian Preference Optimization: Active Learning Loop

1. Data Collection

Solve scalarization subproblems

Obtain approximate Pareto Frontier

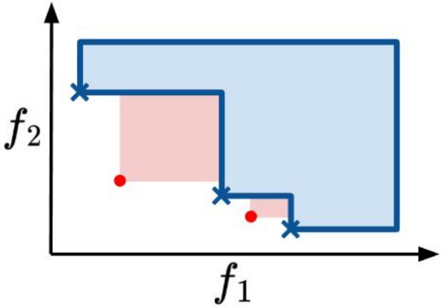


Bayesian Preference Optimization: Active Learning Loop

1. Data Collection

Solve scalarization subproblems

Obtain approximate Pareto Frontier

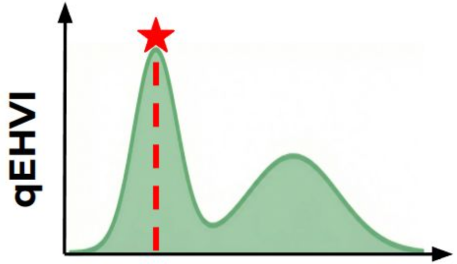


Dominated Region
Hypervolume improvement
Recovered nondominated point (NDP)
Unrecovered NDP

2. Gaussian Process Surrogate Training

$$p(f(\lambda)|\mathcal{D}, \lambda) = \mathcal{N}(\mu_{f(\lambda)}, \sigma_{f(\lambda)})$$

3. Acquisition Optimization



$$\lambda_1 \in [0, 1]$$

$$\lambda_1 : \lambda_1 + \lambda_2 = 1$$

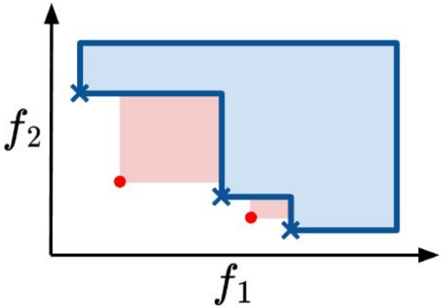
qEHVI peaks when λ_1 = 0.2 preference

Bayesian Preference Optimization: Active Learning Loop

1. Data Collection

Solve scalarization subproblems

Obtain approximate Pareto Frontier

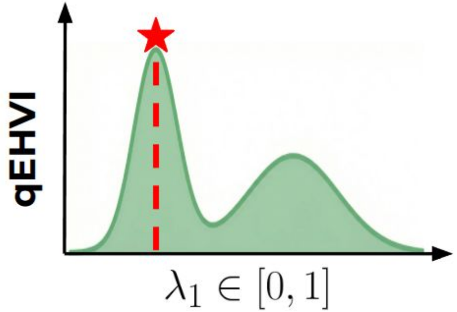


- Dominated Region
- Hypervolume improvement
- Recovered nondominated point (NDP)
- Unrecovered NDP

2. Gaussian Process Surrogate Training

$$p(f(\lambda)|\mathcal{D}, \lambda) = \mathcal{N}(\mu_{f(\lambda)}, \sigma_{f(\lambda)})$$

3. Acquisition Optimization



$$\lambda_1 : \lambda_1 + \lambda_2 = 1$$

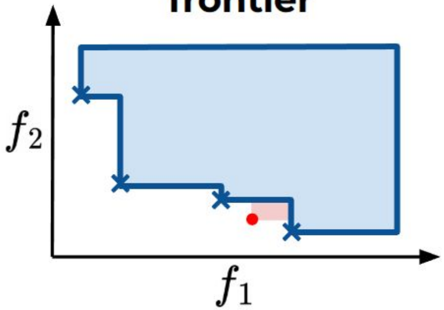
qEHVI peaks when λ_1 = 0.2 preference

4. Solve Subproblem

$$\lambda \longrightarrow \boxed{\text{ATS}} \longrightarrow f(\lambda)$$

Solve ATS with preference optimized using qEHVI

5. Updated Pareto frontier



Recovered NDP that maximizes EHVI

Bayesian Preference Optimization: Active Learning Loop

Algorithm 1: The BPO Framework

Data: Number of initial samples N_{init}, number of main iterations N_{iter}, surrogate model $\mathcal{M}$, acquisition function α

Result: Approximate Pareto frontier $\mathcal{Y}$

```
𝒴 ← ∅, 𝒟 ← ∅
Sample {λ^i}_{i=1}^{N_init} ~ Dirichlet(1)
for k = 1, ..., N_init do
    Solve scalarized problem with parameter λ^k and obtain y^k
    𝒴 ← 𝒴 ∪ {y^k}, 𝒟 ← 𝒟 ∪ {(λ^k, y^k)}
Train surrogate model ℳ using dataset 𝒟
for k = 1, ..., N_iter do
    λ* ← arg max_{λ∈Λ} α(λ | ℳ)
    Solve scalarized problem with parameter λ* and obtain y*
    𝒴 ← 𝒴 ∪ {y*}, 𝒟 ← 𝒟 ∪ {(λ*, y*)}
    Retrain surrogate model ℳ using dataset 𝒟
Return 𝒴
```

Bayesian Preference Optimization: Active Learning Loop

Algorithm 1: The BPO Framework

Data: Number of initial samples N_{init}, number of main iterations N_{iter}, surrogate model $\mathcal{M}$, acquisition function α

Result: Approximate Pareto frontier $\mathcal{Y}$

```
𝒴 ← ∅, 𝒟 ← ∅
Sample {λ^i}_{i=1}^{N_init} ~ Dirichlet(1)
for k = 1, ..., N_init do
    Solve scalarized problem with parameter λ^k and obtain y^k
    𝒴 ← 𝒴 ∪ {y^k}, 𝒟 ← 𝒟 ∪ {(λ^k, y^k)}
Train surrogate model ℳ using dataset 𝒟
for k = 1, ..., N_iter do
    λ* ← arg max_{λ∈Λ} α(λ | ℳ)
    Solve scalarized problem with parameter λ* and obtain y*
    𝒴 ← 𝒴 ∪ {y*}, 𝒟 ← 𝒟 ∪ {(λ*, y*)}
    Retrain surrogate model ℳ using dataset 𝒟
Return 𝒴
```

Dataset generation and initial training

Bayesian Preference Optimization: Active Learning Loop

Algorithm 1: The BPO Framework

Data: Number of initial samples N_{init}, number of main iterations N_{iter}, surrogate model $\mathcal{M}$, acquisition function α

Result: Approximate Pareto frontier $\mathcal{Y}$

$\mathcal{Y} \leftarrow \emptyset, \mathcal{D} \leftarrow \emptyset$

Sample $\{\boldsymbol{\lambda}^i\}_{i=1}^{N_{\text{init}}} \sim \text{Dirichlet}(\mathbf{1})$

for $k = 1, \ldots, N_{\text{init}}$ **do**

- Solve scalarized problem with parameter $\boldsymbol{\lambda}^k$ and obtain $\mathbf{y}^k$
- $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{\mathbf{y}^k\}, \mathcal{D} \leftarrow \mathcal{D} \cup \{(\boldsymbol{\lambda}^k, \mathbf{y}^k)\}$

Train surrogate model $\mathcal{M}$ using dataset $\mathcal{D}$

for $k = 1, \ldots, N_{\text{iter}}$ **do**

- $\boldsymbol{\lambda}^\star \leftarrow \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} \alpha(\boldsymbol{\lambda} \mid \mathcal{M})$
- Solve scalarized problem with parameter $\boldsymbol{\lambda}^\star$ and obtain $\mathbf{y}^\star$
- $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{\mathbf{y}^\star\}, \mathcal{D} \leftarrow \mathcal{D} \cup \{(\boldsymbol{\lambda}^\star, \mathbf{y}^\star)\}$
- Retrain surrogate model $\mathcal{M}$ using dataset $\mathcal{D}$

Return $\mathcal{Y}$

1. Find hypervolume maximizing preference
2. Solve subproblem
3. Update dataset
4. Retrain

Computational Setup

- **Problems**: Multiobjective knapsack and assignment
- 7 different sizes per problem
- 25 problems per size (n: #variables, m: #objectives)
- **Methods**
  - KSA (Exact)
  - NSGA-II (Evolutionary algorithm)
  - ATS (randomly sampled preferences)
  - BPO-100 & BPO-150 (100/150 initial preferences)

n: #variables, m: #objectives

Approximation Quality: BPO vs Baselines

MOKP								
			Hypervolume %					
m	n	Method	Mean.	Std.	Min.	Max.	Iter.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	1250	KSA	64.75	1.65	62.45	68.24	708	706
		NSGA-II	85.57	0.58	84.15	86.80	–	200
		ATS	95.49	0.42	94.18	96.24	217	215
		BPO-100	**96.80**	0.20	96.29	97.23	362	159
		BPO-150	96.73	0.57	94.18	97.28	296	183
6	500	KSA	29.49	2.54	22.89	35.20	26	24
		NSGA-II	65.10	1.85	58.91	69.45	–	200
		ATS	77.75	1.44	73.15	80.60	118	118
		BPO-100	**79.80**	2.83	72.26	84.90	113	107
		BPO-150	78.26	1.82	74.22	84.80	116	115

MOAP								
			Hypervolume %					
m	n	Method	Mean.	Std.	Min.	Max.	Iter.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	200	KSA	74.11	1.19	70.93	76.64	626	625
		NSGA-II	47.68	1.25	45.00	51.86	–	200
		ATS	96.15	1.99	74.78	97.14	185	185
		BPO-100	**97.36**	0.37	95.93	97.80	193	163
		BPO-150	97.01	0.71	95.05	97.88	184	173
7	50	KSA	2.88	0.00	2.88	2.88	26	18
		NSGA-II	8.24	0.49	7.07	9.65	–	200
		ATS	**39.89**	2.26	34.94	46.77	233	233
		BPO-100	37.13	1.51	34.44	42.21	170	159
		BPO-150	38.73	1.70	34.85	43.95	187	182

BPO achieves better hypervolume compared to baselines

n: #variables, m: #objectives

Approximation Quality: BPO vs ATS in Parallel Mode

MOKP							
			Hypervolume %				
m	n	Method	Mean.	Std.	Min.	Max.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	1250	ATS	96.36	0.20	95.94	96.74	1067
		BPO-100	97.09	0.18	96.75	97.46	771
		BPO-150	**97.10**	0.18	96.77	97.47	895
		Oracle	97.15	0.18	96.81	97.51	2180
6	500	ATS	82.02	0.96	79.86	83.81	591
		BPO-100	**84.75**	1.72	80.73	87.53	529
		BPO-150	83.11	1.31	81.45	87.38	577
		Oracle	85.26	1.51	81.66	87.96	761

MOAP							
			Hypervolume %				
m	n	Method	Mean.	Std.	Min.	Max.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	200	ATS	97.49	0.11	97.27	97.68	916
		BPO-100	98.18	0.08	98.01	98.30	740
		BPO-150	**98.20**	0.09	97.98	98.35	831
		Oracle	98.31	0.08	98.14	98.43	1848
7	50	**ATS**	**47.46**	1.82	44.37	52.95	1164
		BPO-100	45.86	1.52	43.59	50.40	773
		BPO-150	46.91	1.59	44.13	51.66	903
		Oracle	49.06	1.92	45.99	54.80	1564

BPO achieves better hypervolume compared to ATS in parallel mode

n: #variables, m: #objectives

Approximation Quality: BPO vs Oracle

MOKP

m	n	Method	Hypervolume % Mean.	Std.	Min.	Max.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	1250	ATS	96.36	0.20	95.94	96.74	1067
		BPO-100	97.09	0.18	96.75	97.46	771
		BPO-150	**97.10**	0.18	96.77	97.47	895
		Oracle	97.15	0.18	96.81	97.51	2180
6	500	ATS	82.02	0.96	79.86	83.81	591
		BPO-100	**84.75**	1.72	80.73	87.53	529
		BPO-150	83.11	1.31	81.45	87.38	577
		Oracle	85.26	1.51	81.66	87.96	761

MOAP

m	n	Method	Hypervolume % Mean.	Std.	Min.	Max.	$\lvert\hat{\mathcal{Y}}_N\rvert$
3	200	ATS	97.49	0.11	97.27	97.68	916
		BPO-100	98.18	0.08	98.01	98.30	740
		BPO-150	**98.20**	0.09	97.98	98.35	831
		Oracle	98.31	0.08	98.14	98.43	1848
7	50	**ATS**	**47.46**	1.82	44.37	52.95	1164
		BPO-100	45.86	1.52	43.59	50.40	773
		BPO-150	46.91	1.59	44.13	51.66	903
		Oracle	49.06	1.92	45.99	54.80	1564

BPO attains hypervolume close to that of the Oracle

Concluding Remarks

- We proposed **BPO**, a principled framework for sample-efficient preference selection aimed at maximizing Pareto frontier quality
- BPO **scales** with number of variables
- BPO is **sample-efficient** compared to baselines and achieves better hypervolume
- **Future work**: Make BPO scalable with the number of objectives

Code + Slides: https://github.com/rahulptel/BPO

References

- Ehrgott, Matthias. Multicriteria optimization. Berlin, Heidelberg: Springer Berlin Heidelberg, 2005.
- Steuer, Ralph E., and Eng-Ung Choo. "An interactive weighted Tchebycheff procedure for multiple objective programming." Mathematical programming 26.3 (1983): 326-344.
- Kirlik, Gokhan, and Serpil Sayın. "A new algorithm for generating all nondominated solutions of multiobjective discrete optimization problems." European Journal of Operational Research 232.3 (2014): 479-488.
- Boland, Natashia, Hadi Charkhgard, and Martin Savelsbergh. "A criterion space search algorithm for biobjective integer programming: The balanced box method." INFORMS Journal on Computing 27.4 (2015): 735-754.
- Mavrotas, George, and Danae Diakoulaki. "A branch and bound algorithm for mixed zero-one multiple objective linear programming." European Journal of Operational Research 107.3 (1998): 530-541.
- Adelgren, Nathan, and Akshay Gupte. "Branch-and-bound for biobjective mixed-integer linear programming." INFORMS Journal on Computing 34.2 (2022): 909-933.
- Halffmann, Pascal, et al. "Exact algorithms for multiobjective linear optimization problems with integer variables: A state of the art survey." Journal of Multi-Criteria Decision Analysis 29.5-6 (2022): 341-363.
- Pal, Aritra, and Hadi Charkhgard. "FPBH: A feasibility pump based heuristic for multi-objective mixed integer linear programming." Computers & Operations Research 112 (2019): 104760.
- An, Duleabom, et al. "A matheuristic for tri-objective binary integer linear programming." Computers & Operations Research 161 (2024): 106397.
- Zhang, Qingfu, and Hui Li. "MOEA/D: A multiobjective evolutionary algorithm based on decomposition." IEEE Transactions on evolutionary computation 11.6 (2007): 712-731.

References

- Li, Kaiwen, Tao Zhang, and Rui Wang. "Deep reinforcement learning for multiobjective optimization." IEEE transactions on cybernetics 51.6 (2020): 3103-3114.
- Lin, Xi, et al. "Pareto set learning for expensive multi-objective optimization." Advances in neural information processing systems 35 (2022): 19231-19247.
- Daulton, Samuel, Maximilian Balandat, and Eytan Bakshy. "Differentiable expected hypervolume improvement for parallel multi-objective Bayesian optimization." Advances in neural information processing systems 33 (2020): 9851-9864.